LeRobot (Free Pascal) Installation Manual

Installing LeRobot on another Mac

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This manual explains how to install LeRobot for Free Pascal on a Mac where it has never been installed. With it, the Turbo Pascal programs of the 1991 book *Premières leçons de programmation en Turbo Pascal* are compiled by Free Pascal and run in the robot window. Once it is done, use the *LeRobot (Free Pascal) User Manual* (`LeRobot_Manual.pdf`) to draw and to run robot programs.

The robot window is the one of the Python version (an ordinary macOS window drawn with tkinter): the Pascal programs send their commands to it. So you need three things: the Free Pascal compiler, a Python with Tk 8.6, and the `LeRobot` folder with its `LeRobot_FreePascal_FR` *and* `LeRobot_Python_FR` folders. No XQuartz is needed. The installation takes about twenty minutes. You need:

- a Mac with a recent macOS (Apple Silicon or Intel);
- an administrator account (the installers ask for the Mac password);
- an Internet connection;
- a copy of the `LeRobot` folder (USB key, AirDrop, cloud drive, ...).

Commands are typed in the **Terminal** application (*Applications* → *Utilities* → *Terminal*). Copy each command in one piece, paste it into the Terminal, and press Return.

Contents

1 The Free Pascal compiler **1**
1.1 Command Line Tools 1
1.2 Free Pascal with Homebrew 2

2 Python with Tk 8.6 **2**

3 The LeRobot folder **2**

4 First launch **3**

5 Troubleshooting **4**

1 The Free Pascal compiler

1.1 Command Line Tools

Free Pascal uses the linker of Apple's Command Line Tools. If Xcode or the Command Line Tools are not installed yet:

```
xcode-select --install
```

and accept the installation in the window that opens (a few minutes).

1.2 Free Pascal with Homebrew

Homebrew (`https://brew.sh`) is the simplest way to install Free Pascal. If it is not installed, install it with the command shown on its home page, then:

```
brew install fpc
```

Check, in a new Terminal window:

```
fpc -iV
```

The answer is the version of Free Pascal (`3.2.2` or later).

Free Pascal can also be installed without Homebrew, with the installer of `https://www.freepascal.org` (*Download*, *macOS*).

Messages to ignore. On recent macOS, Free Pascal 3.2.2 prints linker warnings such as `-macosx_version_min has been renamed` or `-multiply_defined is obsolete`. They are harmless; `robot.sh` hides them.

2 Python with Tk 8.6

The robot window needs Python 3.9 or later *linked to Tk 8.6 or later*. Tk is the library that draws the window.

The `python3` that comes with macOS is not enough. It is linked to Tk 8.5, which no longer works on recent macOS: the robot window does not open and the program hangs. Install Python from python.org, whose installer includes a recent Tk:

1. Open `https://www.python.org/downloads/macos/` and download the *macOS 64-bit universal2 installer* of the latest Python 3 release.

2. Open the downloaded `.pkg` file and follow the installer. At the end, the Finder window it opens can be closed.

Check, in a new Terminal window:

```
/Library/Frameworks/Python.framework/Versions/Current/bin/python3 -c "import tkinter
    ; print(tkinter.TkVersion)"
```

The answer must be `8.6` or more. You do not need to change which `python3` the Terminal uses: when the robot window is started with the macOS `python3`, it notices that its Tk is too old and starts itself again with the python.org Python.

3 The LeRobot folder

Copy the `LeRobot` folder anywhere, for example onto the **Desktop**. The Pascal programs are in its folder `LeRobot_FreePascal_FR`; the folder `LeRobot_Python_FR` must stay next to it, in the same `LeRobot` folder: it draws the robot window. The `LeRobotFreePascal.app` icon must stay in `LeRobot_FreePascal_FR`, next to `robot.sh`. The whole `LeRobot` folder can be moved or renamed later: nothing in it depends on where it is.

Commands. The `LeRobot` folder can be anywhere (on the Desktop, in Documents, ...): everything in it is found relative to it. The commands of this manual are typed in a Terminal window *opened in the `LeRobot` folder*: type `cd` and a space, drag the `LeRobot` folder from the Finder onto the Terminal window, and press Return (or, in the Finder, right-click the folder and choose *Services* → *New Terminal at Folder*).

If the folder comes from another Mac, make the scripts executable again (a copy can lose this) and remove the compiled files of the other Mac (they are made again when needed):

```
cd LeRobot_FreePascal_FR && chmod +x robot.sh verifier.sh outils/*.sh outils/app/*.
    sh && rm -rf _build
```

4 First launch

1. Run a program of the book:

   ```
   cd LeRobot_FreePascal_FR
   ./robot.sh livre/03_programmation/Carre.p
   ```

   The first time, `robot.sh` compiles the robot units (a few seconds), then the program. The robot window (800×700 points) opens in front of the other windows. When the square is drawn, press Esc to close the window.

2. Open the interactive drawing program:

   ```
   ./robot.sh livre/Didactitiels_et_jeux/Dessiner.p
   ```

   Press Return in the window to finish, then `n` not to save.

3. `LeRobotFreePascal.app`, in the `LeRobot_FreePascal_FR` folder: double-click its icon to run Dessiner; drop a `.p` program on its icon to compile and run it. The program runs in a Terminal window, which closes when the program ends normally and stays open if there is an error (of compilation or of execution).

   The first time you open `LeRobotFreePascal.app`, macOS may refuse to open it because it comes from another Mac. Right-click the icon, choose *Open*, then confirm. Allow it to control Terminal and to access the folder of `LeRobot` (Desktop, Documents, ...) when macOS asks.

4. If the folder was copied from another Mac and the application still refuses to open, rebuild it on this Mac:

   ```
   cd LeRobot_FreePascal_FR && outils/app/fabriquer.sh
   ```

5 Troubleshooting

Symptom	Cause	Fix
command not found: fpc	Free Pascal is not installed, or the Terminal was opened before	`brew install fpc` (Section 1); open a new Terminal window
Error: ld: ... or *linker not found*	No Command Line Tools	`xcode-select -install` (Section 1)
Can't find unit Robot1	The robot units are not compiled (folder `_build` removed during compilation)	Run `./robot.sh` again: it compiles them
The program hangs and no window opens	Python linked to Tk 8.5 (the macOS `python3`) and no other Python found	Install Python from python.org (Section 2)
can't open file '.../serveur.py' or *No module named 'LibRobot'*	The folder `LeRobot_Python_FR` is missing or not next to `LeRobot_FreePascal_FR`	Copy the whole `LeRobot` folder (Section 3)
permission denied: ./robot.sh	The scripts lost their executable mark during the copy	`chmod +x robot.sh verifier.sh` (Section 3)
Ce programme doit etre lance par robot.sh printed by a program	The compiled program was started by itself, without the robot window	Use `./robot.sh Program.p` or `LeRobotFreePascal.app`
`LeRobotFreePascal.app` does not open	macOS blocks applications copied from another Mac	Right-click, *Open*; or rebuild it (Section 4)
To use a given Python	—	`LEROBOT_PYTHON=/path/to/python3 ./robot.sh Program.p`